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Stop trying to predict elections only with twitter – There are other data sources and technical issues to be improved

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ABSTRACT

Since the popularization of social media (SM) platforms, researchers have been trying to use their data to predict electoral results. Previous surveys point out that the most used approach is based on volume and sentiment analysis of posts on Twitter. However, they are almost unanimous in presenting that the results are not better than chance. In this context, this study aims to investigate the feasibility of predicting electoral results based only on Twitter, discover the main issues, and draw guidelines for future alternative directions. For this, we reviewed the evolution of election polling and predictions, including the "polling crises" of 1936 and 1948, and their similarities with current approaches. We also built on the official SM platforms' documentation and on our experience collecting and analyzing large-scale data from many SM platforms. Lastly, we analyzed nine reviews on predicting elections with SM data from 2013 to 2021. We observed that, contrary to initial expectations, most of the current research with Twitter has been unable to solve many of the challenges encountered since initial studies, and also shares many of the characteristics of unsuccessful straw polls performed before 1936. We illustrate that by highlighting the impracticability of polling over Twitter due to several biases and technical barriers, the need for external data, the high dependency on the arbitrary decisions of researchers, and the constant change in platforms' scenarios, that may invalidate specific models. Lastly, we indicate some of the possible future directions, such as a focus on creating repeatable processes; the use of SM data as part of statistical models, instead of polling; diversifying the input data sources, including multiple SM platforms and non-SM data such as polls and economic indicators; using machine learning for regression of the vote share, rather than for sentiment analysis; and dealing with the uncertainty of the highly divergent polling results.

1. Introduction

Since the popularization of Twitter[1] and other social media (SM) platforms, such as Facebook and Instagram, researchers have been trying to use data from SM platforms to gauge public opinion and predict elections with SM data. Many surveys, published from 2013 to 2021, summarize the advances in this area (Bilal, Gani, Marjani, & Malik, 2019; Brito, Silva Filho, & Adeodato, 2021; Chauhan, Sharma, & Sikka, 2021; Gayo-Avello, 2013; Kalampokis, Tambouris, & Tarabanis, 2013; Koli, Ahmed, & Manhas, 2019; Kwak & Cho, 2018; O'Leary, 2015; Santos, Bernardini, & Paes, 2021). The previous surveys are unanimous in indicating that the most commonly used approach is based on (Tumasjan, Sprenger, Sandner, & Welpe, 2010) and is firmly based on volume and sentiment analysis of posts on Twitter. However, they are almost unanimous in presenting that this approach is no better than chance, obtaining success rates ranging from 50% to 63%.

Electoral polling and prediction form an old discipline. The first poll similar to modern surveys dates back to 1824 (Smith, 1990). Later, a new form of "straw poll" began, and academics consider that the advent of modern scientific polling appeared in 1936 (Crossley, 1937). Despite its long history, traditional polls have often been criticized, like during the crises of 1936 (Crossley, 1937), 1948 (Mosteller, Hyman, McCarthy, Marks, & Truman, 1949), and 1996 (Mitofsky, 1998). Indeed, there is evidence of contemporaneous biased polls (Fry & Brint, 2017; Kimball, 2019), which have been commonly attributed to unrepresentative sampling, a decline in response rates at the polls, and the effects of social desirability (Kennedy et al., 2018; Sturgis et al., 2018; Wang, Rothschild, Goel, & Gelman, 2015), to cite but a few.

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[1] In the final review phase of this article, Twitter was rebranded to "X"

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On the other hand, over the last decade, we have experienced the advent and popularization of SM platforms, which have established a new communication paradigm that has fundamentally changed both how people communicate and how politicians campaign. Social Media platforms present a new way for politicians to communicate with the electorate, and provide a place where politicians and social researchers may gather, quickly and cheaply, a massive volume of data regarding people's behavior and beliefs. Thus, SM platforms have appeared as a new environment for gathering data regarding public opinion, and are possibly able to compensate for some of the disadvantages of traditional polling.

Within this scenario, Twitter, launched in 2006, is indicated as the first platform to be used as the basis for predicting presidential elections, and was used by (O'Connor, Balasubramanyan, Routledge, & Smith, 2010; Tumasjan et al., 2010) related to elections held in 2008 and 2009. This first SM approach is based on counting the volume of mentions regarding the respective candidates on Twitter and correlating the percentages with the vote share (Tumasjan et al., 2010), and was improved with the sentiment analysis introduced by (O'Connor et al., 2010). In the seminal study above mentioned, Tumasjan stated that "the mere number of tweets mentioning a political party can be considered a plausible reflection of the vote share, and its predictive power even comes close to traditional election polls." More than ten years later, this approach is still the most commonly used, having undergone certain improvements in the sentiment analysis, as presented in recent surveys.

Despite being the most commonly used approach, the use of volume/sentiment analysis on Twitter has also been the most criticized, and began to be challenged just after its launch (Gayo-Avello, 2011; Gayo-Avello, Metaxas, & Mustafaraj, 2011; Metaxas, Mustafaraj, & Gayo-Avello, 2011). Furthermore, Tumasjan's study was directly contested by (Jungherr, Jürgens, & Schoen, 2012), who argued that "the results are contingent on arbitrary choices of the authors" and indicated that simply by including just one more party or collection day the results could be greatly changed. Surprisingly, despite some of the improvements on sentiment analysis, derived from improvements in machine learning research, most of the drawbacks contained in the initial criticisms are still present in the current research, as indicated by (Brito et al., 2021; Chauhan et al., 2021; Santos et al., 2021).

1.1. Problem definition and methodology

In this context, this study aims to investigate the feasibility of predicting electoral results based only on Twitter, particularly presidential elections. For this, it aims to answer the following research question:

What are the main issues with predicting elections based only on Twitter?

To answer this question, we employed a combination of three approaches, consisting of (i) historical analysis, (ii) grey literature analysis, and (iii) a tertiary study of previous systematic reviews and surveys. First, we performed a brief historical review of the evolution of election polling and predictions over time, highlighting some of the problems that have been faced since the beginning of the last century, including the "polling crises" of 1936 and 1948, and their similarities with current approaches. Second, we employed a "grey literature review", a method currently used in software engineering and information systems, based on technical literature and practical expertise (Kamei et al., 2021). Thus, based on the analysis of the official SM platforms' documentation, in addition to the authors' practical experience in developing systems that utilize SM APIs, we have also analyzed the current state, characteristics, and constraints of Twitter and other SM platforms, including the technical constraints on data collection, in order to build a rationale by comparing past and current approaches. Moreover, we performed a tertiary study by analyzing nine surveys related to predicting elections, ranging from 2013 to 2021 (Bilal et al., 2019; Brito et al., 2021; Chauhan et al., 2021; Gayo-Avello, 2013; Kalampokis et al., 2013; Koli et al., 2019; Kwak & Cho, 2018; O'Leary, 2015; Santos et al., 2021), and summarized the results. Thus, critical comparisons and analysis were performed, and have led us to call researchers into predicting elections with SM data to avoid predicting elections only with Twitter, and to present some alternatives. Fig. 1 represents these three approaches.

The remainder of this paper is organized as follows. Section 2 presents the fundamentals related to (a) election polling and predictions, (b) the current state of social media, and (c) the current state of predicting elections with SM data, not only Twitter. Section 3 presents the results of the analyses, including the main issues encountered in the three analyses and focusing on what we consider to be the five main issues. Section 4 presents and discusses possible new directions, followed by Section 5, which discusses the results and limitations of the study. Lastly, Section 6 presents concluding remarks and future work.

2. The fundamentals

2.1. The evolution of election polling and predictions

According to (Smith, 1990), the seminal version of modern opinion surveys was applied in 1824. At that time, the polls were most commonly counts made in many forms: at regular public meetings being held for other purposes, such as Fourth of July celebrations; at meetings specially called to assess the public's presidential leanings, often sponsored by one candidate and as such were really only campaign rallies or informal endorsement meetings; and in "poll books", left out at public places such as taverns over a period of time, for people to write down their preferences.

A new form of straw polls is attributed to the Literary Digest starting in 1916. It was based on sending a huge amount of ballots to citizens by mail and counting the results of respondents. This method was popular until the advent of modern scientific polling in 1936, when prestigious pollsters conducted straw polls of millions of people to indicate a victory for Landon in the U.S. presidential elections, and others, who then started conducting quota-controlled surveys that correctly signposted the victory for Roosevelt (Crossley, 1937)(Squire, 1988).

Motivated by the level of criticism directed towards the polling errors of that particular election, Crossley (Crossley, 1937) classified the polling methods of the time and studied the reliability of each. At that time, polls were either conducted by mail, by personal interview, or a combination of both. Most were performed on huge randomly selected samples, although some were performed on small, scientifically stratified samples. Some used the cumulative method, adding its results together into a single report, and others used sampling at several moments during a campaign. The analysis discovered that polls with better results were conducted wholly or in part by personal interviews, with small samples, and were repeated at frequent intervals. The polls with the worst results, and that used by the Literary Digest, were conducted through the postal service, with a huge sample, and only provided a single complete report accumulated over a period of several months.

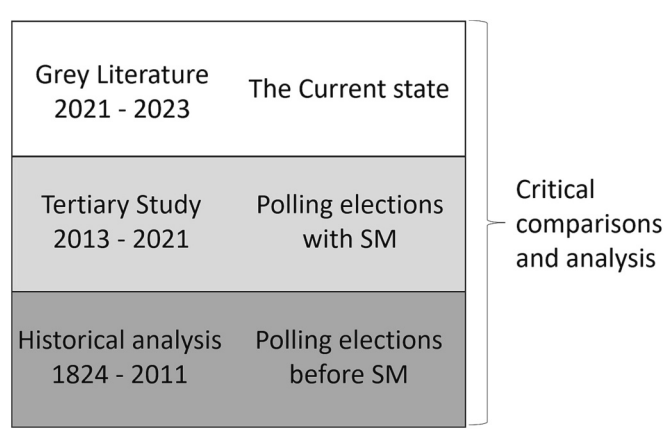

Fig. 1. The three methodological approaches used in this study.

The study highlighted what was considered at the time to be the ideal poll: (i) it needed to be flexible, not based on dated mailing lists, and designed so that it could be readily adjusted if new information became available during its course; (ii) a fairly small sample would work appropriately in all but close states; (iii) the distribution of the sample was of paramount importance; and (iv) it should not be cumulative, but repeated in similar cross-sections at regular intervals to show trends. Many years later, (Squire, 1988) demonstrated that the non-response bias was one of the biggest issues. These principles are still valid today.

Just 12 years after the introduction of straw polls, in 1948, another unexpected result in the U.S. challenged the polling industry yet again. In response to a strong adverse reaction, a distinguished group of social scientists and statisticians mounted an intensive review of the election polling procedures and results. Their report (Mosteller et al., 1949), and the chapter "Measuring the Error" established the measures that have been used ever since to evaluate the accuracy of election polls, but have been barely used by SM researchers.

Since then, election predictions have evolved following the three approaches presented below.

2.1.1. Approaches to election predictions

Three main approaches have been applied for predicting elections: polling, statistical forecasting models, and political stock markets (Hillygus, 2011) (Lewis-Beck, 2005).

In **polling**, researchers directly ask a sample of people for which candidate they intend to vote. Following this, the candidates' vote shares (percentages) from the responses are used to forecast their final vote shares. This approach began with the work of Gallup (Crossley, 1937), and is the most well-known and most commonly-used approach worldwide. It is firmly based on selecting a representative sample of citizens and other design decisions— regarding the mode, timing, sampling method, question formulation, weighting, etc.— so as to enable generalization from a small sample of answers to the behavior of the entire population. Thus, each of these methodological decisions may potentially bias the results, thereby leading to different results from different pollsters. In order to deal with this, the aggregation of many different polls has become popular, and is referred to as polling aggregation or polling the polls (Hillygus, 2011; Jackman, 2005). Aggregation finds theoretical support in ensemble techniques for decision making in Artificial Intelligence (AI), more particularly on Bagging Predictors (Breiman, 1996). In a very summarized explanation, the Bootsttrapping AGGregatING (BAGGING, for short) algorithm consists of combining the results of n predictors to synthetize their results. These predictors of the same nature (usually Decision Trees) are trained with different data sets obtained by Bootstrapping, a well stablisehd statistical sampling technique. The final result will present much less variation and biases of some types tend to be reduced by the compensation of each individual result through the aggregation process.

Statistical forecasting, also called macroeconomic models, began to appear around 1980 (Fair, 1978; Sigelman, 1979). At the core of these models is the assumption that the vote share is a function of other indicators, such as government performance, economic performance, or economic growth. These statistical models may also take into account polling results, such as incumbent popularity. A general theory is that voters reelect incumbents during good (economic) times and do the opposite during bad times (Fiorina, 1978). Thus, the debate largely regards which economic indicator to be used and also regards the inclusion of other variables in the model, such as polling numbers (Abramowitz, 2004; Lewis-Beck, 2005).

The third approach, the **political stock market**, considers how traders invest money in candidates running for office. In this market, people buy and sell candidate futures based on whom they judge as being more likely to win, and the investment share of any one candidate provides a vote forecast for the candidate (Rhode & Strumpf, 2004). This approach may be considered a specialized polling approach since it polls the opinion of traders. Although traders are not a representative sample, they are just assumed to be making informed judgments and have the confidence to wager their money on it.

2.1.2. Measuring prediction accuracy

After the 1948 polling crisis, the report produced by Mosteller et al. (Mosteller et al., 1949) established eight different metrics for assessing polling accuracy. Two of these measures, the "Mosteller Measure 3" and the "Mosteller Measure 5", have been widely used ever since in order to evaluate the accuracy of elections polls (Hillygus, 2011; Jennings & Wlezien, 2018).

The "Mosteller Measure 3" is the average deviation of the candidates, each measuring the absolute value of the percentual difference between the predicted and the electoral (ballots) vote shares. In statistics, it is widely known as the mean absolute error (MAE). The "Mosteller Measure 5", which we call the absolute error on the margin (AEOM), is the absolute value of the difference between the margin separating the two leading candidates in the poll and the difference in the actual vote.

Other metrics may also be used, such as the "Mosteller Measure 4", which measures the percentage error and is known in statistics as the mean absolute percentage error (MAPE), and the root mean squared error (RMSE), which penalizes outliers.

With emphasis on the MAE, these metrics were used by (Jennings & Wlezien, 2018) to assess prediction errors in pre-election polls. The analysis drew on more than 30,000 national polls from 351 general elections in 45 countries between 1942 and 2017. The study observed that, contrary to conventional wisdom, the recent performance of polls has not been out of the ordinary. By calculating the estimated average poll during the week prior to elections, they discovered that the MAE had almost no change, from 2.1% during the 1940s and 1950s, to 2.0% since 2000. They also reported that errors were higher in presidential elections (an average of 2.7 percentage points) than in legislative elections (1.8 percentage points). Lastly, as expected from the cone of uncertainty used in forecasting applications (Spiegelhalter, Pearson, & Short, 2011), they reported that errors decreased as the elections approached.

These metrics should be used in future studies regarding predicting elections with SM data, instead of just indicating the winner or claiming that the predictions were close to the official vote share, as so often occurs in many studies as already been criticized (Brito et al., 2021).

2.2. The current state of social media

Currently, more than half (59%) of the global population actively uses SM platforms (Kemp, 2023). However, these platforms are new: Twitter debuted in 2006, Facebook was launched for public access also in 2006, and Instagram emerged in 2010. This scenario is also constantly changing: Kuaishou, also known as Kwai, was launched in 2011, and TikTok in 2016 which has already become very popular (Xu, Yan, & Zhang, 2019). Fig. 2 shows a projection of users in the world's most popular SM platforms in 2023. The data reveals that Facebook, Youtube, Whatsapp, Instagram, WeChat, TikTok, and FB Messenger are the most widely used platforms.

These platforms may be classified into two groups. In conversation platforms, the main feature consists of allowing direct conversation between two (or a group of) people. Examples are Whatsapp, Facebook Messenger and Telegram. In newsfeed platforms, the main features consist of a pair of posts/newsfeed. People are encouraged to post contents, which may contain text, photos, audio, video, and links. The contents are shown to their connections in a newsfeed format, and viewers may interact with them, usually either by clicking "like" (signaling that he/she liked the content), by commenting on the post, or by sharing it with his/her own connections, thereby amplifying the reach of the message.

2.2.1. Newsfeed algorithms and the bubble effect

As a matter of privacy, direct conversation platforms do not usually

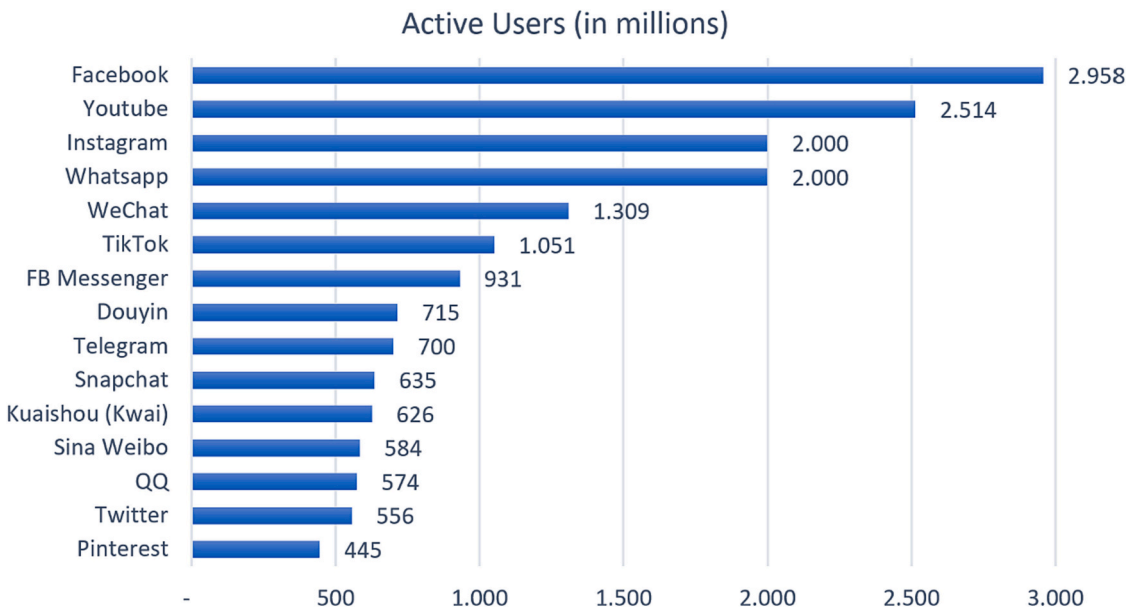

Fig. 2. Projection of the most commonly used SM platforms worldwide in 2023, according to (Kemp, 2023).

allow data collection by third parties. Thus, the newsfeed platforms are most suited for studies on predictions. On these platforms, people may connect to, or just follow, either a few or thousands of other accounts. Thus, since it is impracticable to show users all the posts of all their connections at once, platforms use optimization algorithms to choose which content to show on their news feed, in order to maximize their chance of interaction.

Since 2017, the Twitter algorithm has used deep learning to build user timelines. They explain that "Right after gathering all Tweets, each is scored by a relevance model. The model's score predicts how interesting and engaging a Tweet would be specifically to you. A set of highest-scoring Tweets is then shown at the top of your timeline, with the remainder shown directly below"(Koumchatzky & Andryeyev, 2017). They also detail that the algorithm takes into account the following measures: (i) The Tweet itself: its recency, presence of image or video, total interactions (e.g. number of Retweets or likes); (ii) The Tweet's author: the user's past interactions with this author, the strength of the user's connection to him/her, the origin of the relationship; and (iii) You: Tweets that the user found engaging in the past, and how often and how heavily the user uses Twitter.

The algorithms of other platforms are similar. Facebook (Lars Backstrom, and The Facebook, 2013) prioritizes showing popular content to users and content that corresponds to what they have previously liked. Similarly, Youtube explains that the home screen is based on two items: performance, or how well a video has engaged and satisfied similar viewers, and personalization, based on a viewer's watch and search history (Youtube Inc, 2017).

These algorithms help users by showing them what they probably want to see. However, this approach has received much criticism because it leads to filter bubbles. As users are only presented with content that they and their connections already like and are familiar with, possible new viewpoints and opposite ideas are filtered out, leading to a false sense of unanimity. Extensive research into the downside of this scenario is currently being performed by social scientists (Flaxman, Goel, & Rao, 2016; Spohr, 2017).

Newsfeed SM platforms may be viewed as places where it is possible to gather public opinion, especially if we consider that half the world's population is currently using them. Thus, SM allows researchers to gather a huge amount of data from an unpredictable number of people at a low cost (Jungherr, Schoen, Posegga, & Jürgens, 2017). However, collecting this data may also be a challenge. Next, we overview the official data collection methods from Twitter.

2.2.2. *Social media data gathering*

The publishing of data by SM platforms is an important part of the entire ecosystem of SM. This publishing allows the development of third-party applications, from social games to entire marketing platforms. It also allows social studies. However, publishing this data has become a sensitive topic, and several mechanisms are under discussion in order to protect user rights, especially after the Cambridge Analytica scandal (Isaak & Hanna, 2018). In this particular episode, Cambridge Analytica used data analytics to sway the electorate, relying on the participation of SM platform users through their psychological manipulations.

Twitter and other newsfeed platforms have well-defined methods for publishing data that has been collected from them through the use of application programming interfaces (API's). The prime API version of Twitter until the end of the year 2021 was v1.1 (Twitter Inc, 2022), when they officially brought in the v2 API as their primary API.[2] Due to the recency of the new version (v2) at the time of writing this paper, we have focused on explaining v1.1, the version used by previously published works. The novelties of v2 API and the current state of changes are discussed in Section 3.4.

Twitter API allows the search for users and gathering their metadata, such as name, picture and biography, and all his/her posts. It also allows an open search on the platform based on keywords. This may be served in two ways: as a traditional query request with an (almost) immediate response and as streaming connections to serve APPs with real-time tweets.

However, due to the huge number of tweets performed every second and the amount of information that may be retrieved from the API, the platform presents some limits. First, for basic, free access, the Twitter platform does not guarantee that all tweets will be retrieved. In fact, it simply returns a sample of tweets, as the official documentation states, "The Twitter Search API searches against a sampling of recent Tweets published in the past 7 days."[3] There are also rate limits. For the search

[2] The announcement was made through their official blog: https://blog.twitter.com/developer/en_us/topics/tools/2021/build-whats-next-with-the-new-twitter-developer-platform

[3] https://developer.twitter.com/en/docs/twitter-api/v1/tweets/search/overview

API, there is a limit of 100 tweets per request and 180 requests per user in a window of 15 min. Hence, a maximum of 18.000 tweets per user may be collected in the time window, which could be sufficient for searches with few results, but not for presidential elections in populous countries. Even if this were sufficient, Twitter does not guarantee completeness, and the researcher would never be aware of how many Tweets had not been included. One alternative for achieving completeness is to use Twitter premium API v1.1. This version is a paid API, which allows searches on the full archive of Twitter. However, in addition to prohibitive costs, the rate limits also prevent collecting the huge amount of Tweets posted during electoral periods in populous countries.

Despite these drawbacks, we nonetheless believe that Twitter is the most commonly used platform for predicting elections due to a particular feature on this API: the open search capabilities. With this, researchers are able to search just for "Biden", "Merkel" or "Putin" and receive Tweets in which they are mentioned. This functionality is not present in this form on the APIs of other platforms, such as Facebook and Instagram (Facebook Inc, 2022), which are more focused on the timelines or, at most, open search on hashtags.

It should be noted that the new version of the Twitter API launched in November 2021 allows a new level of access for qualified researchers. It will be discussed in Section 3.4.

2.3. A summary of 12 years of research on predicting elections with SM data

We have found in the literature nine surveys on predicting elections with SM data, not only Twitter, performed between 2013 and 2021, which are described below. Table 1 presents a summary of the main points of each survey.

In 2013, Kalampokis et al. (Kalampokis et al., 2013) presented a systematic review that aimed at understanding the predictive power of SM, not limited only to the electoral context. By analyzing 52 studies, 12 regarding election predictions, they identified that the main approaches were based on volume, sentiment, and user profiling. Moreover, the application of predictive analysis using linear regression was also identified, although not in studies related to the political context. They also verified that 40% of the studies that had used sentiment-related variables were unsuccessful and that this percentage increased to 65% with lexicon-based approaches. Lastly, they emphasized a lack of assessment for predictive analytics and controversial results of electoral predicting studies.

In the same year, Gayo-Avello (Gayo-Avello, 2013) presented a study that we consider to be the first review that specifically centered on predicting elections with SM, and focused on Twitter. By analyzing 10 previous studies from 2010 to 2013, he concluded that "the presumed predictive power regarding electoral prediction has been somewhat exaggerated." Moreover, as in (Kalampokis et al., 2013), he identified volume and sentiment analysis as the main approach as well as the need to use more up-to-date methods for sentiment analysis. He also expanded the list of challenges, such as the dependency of arbitrary decisions made by researchers regarding keywords, parties, candidates and selecting the data collection period, and problems related to Twitter, such as demographic and self-selection bias, including bias related to spam, misleading propaganda and astroturfing. He ended the study by mentioning that regression models would be a future direction.

In 2015, the study performed by O'Leary (O'Leary, 2015) presented a general overview of the main approaches for prediction using Twitter in many different domains, and briefly described a few studies related to election predictions. His analysis highlighted the controversy concerning positive and negative results regarding whether it is possible to use the number of tweets mentioning a candidate or party to predict elections. In 2018, Kwak (Kwak & Cho, 2018) presented the results of a survey, including 69 papers, which supported the argument that SM can be used for understanding political agendas, rather than election forecasting. More recent studies, (Koli et al., 2019) and (Bilal et al., 2019), have presented limited nonsystematic surveys, both analyzing 13 papers, and have added some arguments to the original review by Gayo-Avello (Gayo-Avello, 2013). Koli (Koli et al., 2019) argued that prediction using Twitter may have better results in developed countries than in developing countries, due to a higher literacy rate and wider internet access. In addition, Bilal (Bilal et al., 2019) considered the challenges of sentiment analysis in languages other than English.

In 2021, three surveys summarized these initiatives. Chauhan et al. (Chauhan et al., 2021), surveyed 38 papers, Brito et al. (Brito et al., 2021) performed a systematic review of 83 relevant studies, and Santos et al. (Santos et al., 2021) analyzed 53 studies.

Chauhan et al. (Chauhan et al., 2021) began from the premise that forecasting election results is an application of sentiment analysis or opinion mining by gauging the mood of the public through Twitter. They found the highest success rate, 63%. Their major findings were in line with previous studies: most used Twitter as a corpus for prediction, researchers focused on sentiment-oriented content analysis and the most commonly used sentiment analysis was the traditional lexicon-based approach.

In a broader study, while Brito et al. (Brito et al., 2021) agreed that the volume/sentiment analysis on Twitter is the most commonly used approach, they also added that this approach presents a low success rate (just 55% of the 64 studies that used this approach achieved success). They also summarized the main challenges into four categories: process, sampling, modeling and performance evaluation. From these, we are able to highlight the challenges in (i) defining repeatable processes; (ii) the no representativity of sampling over Twitter; (iii) the choices of arbitrary data collection, such as keywords and the period of data collection; (iv) the high susceptibility to volume manipulation and the presence of a natural bias in the differences in supporter behavior on SM platforms; (v) the difficulties in using data from multiple networks besides Twitter; and (vi) the need to use ML models, not only for sentiment analysis but also for predictions.

Lastly, the most recent survey (Santos et al., 2021) agreed with (Brito et al., 2021) on most points. They presented a similar low success rate (54% of the 28 studies using counting-based approaches on Twitter) and emphasized the need for processes that are not dependent on arbitrary choices and for models not susceptible to volume manipulation. They also indicated new challenges involved in using ML models for

Table 1
Surveyed papers.

Author	Year	Domain	Platforms	Studies	Systematic	Success on Twitter
Kalampokis et al.	2013	Many	Many	52	Yes	60%
Gayo-Avello	2013	Elections	Twitter	10	No	50%
O'Leary	2015	Many	Twitter	27	No	–
Kwak & Cho	2018	Elections	Many	69	No	–
Koli et al.	2019	Elections	Twitter	13	No	Dev. countries
Bilal et al.	2019	Elections	Many	13	No	–
Chauhan et al.	2021	Elections	Twitter	38	No	63%
Brito et al.	2021	Elections	Many	83	Yes	55%
Santos et al.	2021	Elections	Many	53	Yes	54%

sentiment analysis on Twitter, especially those related to data labeling for training the models during the short period of electoral campaigns. Also, they point the lack of availability of domain (labeled) datasets for pre-training the models.

3. The Main issues with predicting elections based only on twitter

The reviews presented in Section 2.3 shows that the success rate of the approaches based only on Twitter, considering from the first surveys of 2013 until the last ones in 2021, ranges from 50% to 63%, with an average of 56%. Indeed, it can be considered no better chance and has not improved over time. Thus, we can consider that predicting elections with Twitter is a very challenging, if not impractical, task.

In order to answer the research question, "*what are the main issues with predicting elections based only on Twitter?*", we analyzed together the results of the three analyzes (i) the surveys' results and discussion, (ii) the viewpoint of historical knowledge on the poll industry, and (iii) the current state of SM platforms. Thus, we emphasize five main issues.

First, we believe that it is not feasible to poll over Twitter posts, and that statistical forecasting should be used. Second, for statistical forecasting, other sources are needed in addition to Twitter, particularly offline sources. Third, in practice, efficient methods for filtering Twitter posts may only be clearly defined after the elections, since they depend on several arbitrary decisions. Fourth, the SM and Twitter landscape constantly evolves, and building models that focus solely on one specific platform may suddenly become obsolete. Lastly, we have identified that contemporaneous research is repeating the same errors of straw polls performed before 1936.

3.1. It is not feasible to poll over twitter posts

The main assumption of studies based on volume on Twitter is to use this platform to poll the opinion of citizens. Thus, if X% of posts are related to a candidate, he/she will probably obtain X% of the vote share. The approach is enhanced by the use of sentiment analysis to identify whether the opinion of the user is in favor or against the candidate, and some minor improvements (such as avoiding duplications) are usually performed.

Three main reasons lead us to believe that we can not (or at least, should not) poll over Twitter posts.

Twitter offers a very biased sample of the voters: Polling methods are firmly based on selecting a representative sample of respondents to enable extrapolation from a few answers to the entire population. However, it is clear that Twitter is not a representative sample of the entire population. In fact, it is a strongly biased sample. For example, a report published in 2019 observed that Twitter users in the U.S. were younger, likelier to be identified as Democrats, more highly educated, and had higher incomes than U.S. adults overall (Pew Research Center, 2019). These data do not reflect world or U.S. demographics. Moreover, this scenario has probably changed since the report, as the popularization of TikTok has attracted a large audience of young people (Xu et al., 2019).

Some may argue that the large amount of data collected off the platform from a huge quantity of people may compensate this bias. However, even though more than 50% of the population use SM, Twitter is not even among the top 10 of the most commonly used platforms, as previously presented in Fig. 2. Thus, because of this, other SM platforms should also be considered.

A direct correlation can not be guaranteed: Although recent research has reported a correlation between SM performance and electoral performance in a specific context [48], it should not be assumed that this occurs in all elections worldwide. For example, in the famous response to the Tumasjan's seminal study, Jungherr (Jungherr et al., 2012) found that the party which obtained the most mentions on Twitter obtained only 2% of the vote. This may occur on Twitter and SM platforms in general, because of many reasons, such as the use of automated software, known as BOTs (Bessi & Ferrara, 2016) spammers, paid propaganda, astroturfing, coordinated organization of supporters on SM, or even the natural differences between the behavior of users (Mustafaraj, Finn, Whitlock, & Metaxas, 2011).

Twitter technical reasons: As presented in Section 2.2.2, the Twitter API clearly states that "The Twitter Search API searches against a sampling of recent Tweets published in the past 7 days", and that "This means that some Tweets and users may be missing from search results. If you want to match for completeness you should consider the premium or enterprise search APIs". Even considering the use of premium or enterprise APIs, the costs and rate limits are highly prohibitive for using in populous countries with millions of posts related to elections every hour. This reason may have been softened by the launch of the new Twitter API in November 2021, allowing researchers to gather data with lower limitations. However, it will be limited to researchers authorized by Twitter, and still needs to be validated by the research community.

3.2. To use statistical forecasting, other sources and historical data are needed

Due to the fundamental challenges of using Twitter as a polling strategy, we argue that the data collected on this platform, and at SM platforms in general, should be used in a statistical forecasting approach. As presented in Section 2.1.1, at the core of these models is the assumption that vote share is a function of different indicators, such as government performance, economic performance and growth, and also polling results. An example from the U.S. presidential elections was offered by (Lewis-Beck, 2005) and is presented below:

$$V = 37.31 + 0.28P + 1.32E + e.$$

where V = presidential party share in the election; P = presidential popularity, in the July Gallup Poll of the election year; E = percentage growth in the real GNP over the first two quarters of the election year; and e = error.

Statistical forecasting is also strongly based on historical data, used for the adjustment of the function's parameters. Thus, in a possible setup, Twitter or other SM data may be used as part of a broader equation. However, since Twitter is a new phenomenon and presidential elections are usually held at 4 or 5 year intervals, the lack of historical data regarding its use may be prohibitive for making accurate predictions even in this setup.

3.3. Methods for data gathering mentions may only be clearly defined after the elections

Starting from initial papers challenging the use of Twitter for predicting elections (Gayo-Avello et al., 2011; Jungherr et al., 2012; Metaxas et al., 2011), and including all reviews presented in Section 2.3, they all agree on one point: predictions are based on arbitrary choices for data collection. These choices include, but are not limited to, two main choices: the time and duration of data collection and the keywords for filtering the posts.

The literature states that the time of data collection varies from the last day or week before elections to some months before. The duration of data collection also varies from one day to more than 90 days of collection. These decisions are barely justified, and a baseline data collection period has not yet been found.

The selection of keywords for filtering is also challenging. For example, some candidates have "common" names, and a search for their names may include many results unrelated to elections. Also, candidates sometimes adopt nicknames that people may prefer to mention, or new nicknames may just naturally appear and become adopted by people during the campaign. Lastly, misspelling or abbreviations are also very common. Thus, the correct selection and pruning of these keywords may

be not practical for performing in real time during the campaigns.

3.4. Constant changes in external and internal scenarios

The SM landscape is constantly evolving. In 2016, the year TikTok was launched, it was estimated that 31% of the world's population used SM. Facebook was the most used platform and Twitter ranked ninth, including both wall-based and direct conversation models (Kemp, 2016). However, only seven years later, at the beginning of 2023, the share of active social media users had increased to 59% of the global population. While Facebook remained the most widely used platform, TikTok emerged as the sixth most popular, and Twitter dropped to the fourteenth position (Kemp, 2023). Therefore, building models that focus solely on one specific platform may introduce a strong bias, especially if that platform is losing its share of active users.

Regarding the internal scenario, the recent release of the Twitter API (v2) introduced a new level of access for qualified researchers, which promises authorized researchers to have access to the full Twitter basis, mitigating some problems pointed out in Section 2.3.

On the other hand, following Elon Musk's acquisition of Twitter in 2022 (Jia & Xu, 2022), several changes were performed to the platform, leading to uncertainty for all its stakeholders, from advertisers to developers and developers researchers (Bebchuk, Kastiel, & Toniolo, 2023). For example, on February 2, 2023, the official Twitter account for communication with developers suddenly published that "Starting February 9, we will no longer support free access to the Twitter API, both v2 and v1.1. A paid basic tier will be available instead,"[4] surprising of all the ecosystem. The announcement of this change, made only a week prior, holds the potential to disrupt thousands of services that rely on collecting API data, including research and commercial projects. On February 08, just a few hours before the deadline, they extended it to February 13,[5] when they extended again "by a few more days."[6] Finally, on February 17, they published a generic message[7] mentioning "over the next few weeks." As a result, at the time of this manuscript writing, researchers were uncertain about how they would be able to access the Twitter API in the coming weeks.

3.5. We are repeating the same errors of straw polls performed before 1936

Current research on predicting elections with Twitter is, in many ways, similar to straw polls that started in 1824, which led to the polls crisis of 1936 (Crossley, 1937).

Collecting posts on Twitter may be seen as a combination of the three main modes of data collection which began in 1824: (i) it is a "poll book", which people may voluntarily just "sign", by publishing his/her opinion regarding a candidate; (ii) it is a public meeting, where a massive quantity of people gather together for purposes other than elections and eventually talk about elections; and (iii) it usually contains informal endorsement meetings, especially called for this purpose, when candidates or parties post and prompt their supporters to like, share or act in an organized manner.

Moreover, it is also similar to the Literary Digest of 1936, based on sampling a huge number of people in certain ways. The biased sample used for the poll has been blamed as the source of the flawed result, and a similar bias occurs when using Twitter, as previously presented. In addition, the non-response bias is also indicated as an important component of errors (Squire, 1988). This non-response bias is strongly present on Twitter, due to a minority of people being responsible for the majority of posts (Mustafaraj et al., 2011), and it is reasonable to suppose that most citizens prefer not to show their political leaning in a Twitter post.

In addition, most current research uses the cumulative method, adding its results together into a single report. This approach is the opposite of the 1936 advice (Crossley, 1937). At that time, it was found that polls with better results were conducted and repeated at frequent intervals, discarding previous results. On the other hand, polls with the worst results, and the one used by the Literary Digest, only provided one complete report accumulated over the collection period, such as in contemporaneous Twitter-based approaches.

4. Possible directions

This section presents some of the possible directions for research on predicting elections with SM data. Since there are many possible directions, the intention is not to provide an exhaustive list, but rather to contribute with some guidance.

4.1. Create a repeatable process for the future rather than a model of the past

Most studies are only applicable and applied indeed in one electoral context and are performed only after the elections are concluded. Developing a solution endowed with predictive capability, only to employ it as a descriptive tool, would be deemed futile. Also, as previously discussed, most of them are subject to arbitrary decisions made by researchers that may involuntarily bias the results. Thus, the first advice is that researchers should focus on defining general and repeatable processes, which have a true predictive power of predicting elections in multiple scenarios, and assessing the processes with several case studies. The assessment should also use the consolidated metrics of the polls domain, as presented in Section 2.1.2.

4.2. Use SM as a part of a statistical model, not polling

As discussed in Section 3.2, for a number of different reasons, while it is practically impossible to use Twitter for polling, it may be used as a part of statistical forecasting. Thus, the use of statistical models may be promising. For example, statistical time series models (De Gooijer & Hyndman, 2006) such as exponential smoothing, auto-regressive moving average (ARMA), auto-regressive integrated moving average (ARIMA), and Kalman filter, as well as nonlinear models, such as Artificial Neural Networks (ANN) (Hassoun, 2010) may be used. One possible use is the daily estimations of the tendencies of candidates, as to whether one candidate is more likely to increase or decrease the vote share based on the perceptions gathered from SM data. Lastly, these models may be combined with other data sources, as presented below.

4.3. Diversify the input data sources, including multiple SM platforms and non-SM data

Researchers should consider the constant changes both in the external scenario, such as the popularization of new SM platforms, and in the internal scenario, such as the sudden changes of policies on Twitter. The change of internal scenario may occur in any SM at any time, not only on Twitter. Thus, models should not be focused on only one platform, and other ones, such as Facebook, Instagram, or TikTok, should also be considered for use in the models. Indeed, recent studies conducted in different electoral contexts found different yet strong correlations between SM performance and electoral results on Twitter, Facebook, and Instagram (Brito & Adeodato, 2022) and even on TikTok (Lima, Santana, Correa, & Brito, 2023). Thus, mixing the platforms, each with a specific bias regarding its users, may be used in a similar manner to traditional polling aggregation (Jackman, 2005), inheriting some of its benefits. Some insights from scientific domains such as neuroscience (Buonomano, 2011) and ensembles of stacked classifiers (Wolpert,

[4] https://twitter.com/TwitterDev/status/1621026986784337922
[5] https://twitter.com/TwitterDev/status/1623467615539859456
[6] https://twitter.com/TwitterDev/status/1625234161010343941
[7] https://twitter.com/TwitterDev/status/1626732269174943745

1992), which have analyzed multiview learning, might be useful for merging databases from different sources with electoral purposes.

Similarly, by using statistical methods, it is expected that additional data sources, other than SM data, should be included. For instance, using historical data from past elections, economic indicators, or traditional polls, as already used in other statistical models, may be promising. We highlight the potential benefits of using traditional poll data, which may be considered the best available approximation to actual public opinion. Their main disadvantage is that they demand long periods of time and high costs in order to perform. Thus, in many countries, they are performed at considerable intervals. However, by combining SM data with poll data, as in (Brito & Adeodato, 2023) new models can benefit from eventually receiving "imprecise ground truth" from the polls and collecting daily data from SM platforms. Thus, daily predictions may be performed.

4.4. *If machine learning is used, use it for regression rather than sentiment analysis*

As the current use of Twitter is mostly for polling, the focus of using ML models is more on improving sentiment analysis than on improving the prediction model. However, the vote share prediction may clearly be considered a regression task, when methods are designed to predict a continuous numeric output where an order relation is defined. There are several different approaches for regression (Fernández-Delgado et al., 2019), most of them suitable for this domain, but not as yet studied within this context. One possible direct application of ML regression as the basic model for predictions is to use sources from multiple SM platforms as features and poll data as labeled data. Thus, the model may be trained with poll data and used for performing daily predictions.

4.5. *Use polls as probabilistic targets rather than deterministic*

Methods using polls as ground-truth face an old and important issue in the field of AI under the supervised learning paradigm, since the recorded outcomes (desired results) are highly impacted by noise. In general, noise can arise from inherent problem complexity, inaccuracies in the measurement device, or poor data recording and storage practices. In the case of polls, estimation errors are an inherent source of noise, as indicated in all disclosures of poll results. Indeed, even when conducted on the same day, different pollsters produce varying results for similar pools.

AI has developed some methods to reduce the negative effects of this type of uncertainty, mostly based on noise estimation (Nettleton, Orriols-Puig, & Fornells, 2010) (Vabalas, Gowen, Poliakoff, & Casson, 2019). Instead of being a problem for modeling in election prediction, the error distribution inherent to the polling process might improve model quality. Thus, despite using poll results as the target variable in a deterministic fashion, the target variable may be dynamically generated according to the error distribution of the polls, at each iteration towards the convergence of the algorithm. Thus, this procedure can not only increase the performance of predictors, but also reduces the high risk of overfitting in such small datasets.

Table 2 summarizes all discussed main issues and possible directions.

Table 2
Summary of discussed main issues and possible directions.

Main issues
It is not feasible to poll over Twitter posts
To use statistical forecasting, other sources and historical data are needed
Methods for data gathering mentions may only be clearly defined after the elections
Constant changes in external and internal scenarios
We are repeating the same errors of straw polls performed before 1936
Possible directions
Create a repeatable process for the future rather than a model of the past
Use SM as a part of a statistical model, not polling
Diversify the input data sources, including multiple SM platforms and non-SM data
If Machine Learning is used, use it for regression rather than sentiment analysis
Use polls as probabilistic targets rather than deterministic

5. Results discussion

This paper presented the main issues with predicting elections based only on Twitter. Some of them are not a novelty and have been presented since the initial studies (Gayo-Avello, 2013; Kalampokis et al., 2013). Indeed, some of the issues presented in Section 3 are present both in those studies and also in the new surveys (Brito et al., 2021; Chauhan et al., 2021; Santos et al., 2021). However, as a tertiary study, the assessment and conclusions that many of the initial problems remain unresolved, and the success rates have not increased, is an important contribution.

An additional contribution is the presentation and discussion of the evolution of election polling and prediction, which is not present in other studies in this area. The lack of knowledge on this subject may be the cause of repeating the errors already overcome by the poll industry. Also, the analysis of the practical situation of Twitter, and the provided guidance, are additional contributions.

We can highlight two limitations of this study. First, the selection process of the surveys that served as a basis for the analysis was non-systematic. Thus, valuable studies may be missing. Second, despite doing their best to provide an extensive list of possible directions, the authors acknowledge that the list is far from being exhaustive, and many possible and different directions may appear in a new future.

Lastly, this study focuses on Twitter and the results can not be all generalizable for al SM platforms. If we consider other popular platforms, such as Facebook, Instagram, or the recent TikTok, we can find some similarities and differences regarding the conclusions. As the main difference, the open search capability of Twitter is not present on these other platforms. Thus, the capability of trying to poll over mentions, and all related issues, especially those presented in sections 3.1, 3.3, and 3.5, is not present in the majority of other platforms. On the other hand, all possible directions may be applied regardless of the platform under study.

6. Concluding remarks

Strategies based on volume and sentiment analysis of Twitter posts are, by far, the most commonly found in the literature when the objective is predicting elections based on SM data. However, this approach also receives the most criticism, has its results constantly challenged, and its success rate may be considered no better than chance.

This study has aimed to investigate the feasibility of predicting electoral results based only on Twitter, especially in presidential elections. For this, it answered the research question "*What are the main issues with predicting elections based only on Twitter?*". Thus, we have analyzed the evolution of polling since its early roots in 1824, the contemporaneous scenario of Twitter and SM platforms, and several surveys on predicting elections with SM data performed between 2013 and 2021. Lastly, we have highlighted the major identified problems and indicated some possible directions.

As the main problem, we pointed that it is not feasible to poll over Twitter posts. This is due to several reasons: it is not compliant with the fundamental premise of polling, which is based on selecting a representative sample of voting citizens, and there is strong evidence that Twitter polled data can not be extrapolated to the entire population, plus there are also technical barriers on Twitter API. One possible alternative, statistical forecasting, requires other sources and historical data. Moreover, most methods may only be clearly defined after the elections.

Additionally, it should be avoided to be dependent on a single platform due to the bias regarding the users of the platform, the constantly changing SM landscape (for example, Twitter currently is the fourteenth

most used platform while the newer TikTok is the sixth most used), and the potential changes that may occur to a platform, such as what happened with Twitter after its acquisition in 2022.

Surprisingly, researchers are repeating the same errors of straw polls performed before 1936. Despite digitalization, current approaches may be considered a mix of "poll books" and massive data gathering. Moreover, they do not consider the critical non-response bias, and are based on a cumulative method, among other similarities.

Lastly, the study has presented the authors' viewpoint regarding the future directions of predicting elections using SM data. The main recommendations derived from this study are:

- To create repeatable processes for the future, rather than a model of the past;
- To use SM as a part of a statistical model, not polling;
- To diversify input data sources, including multiple SM platforms and non-SM data, specially polls;
- If Machine Learning is used, use it for regression rather than sentiment analysis; and
- The ML supervised learning approach should benefit from the error inherent to polls.

The results of this study contribute to the research field by providing the academic community and practitioners with a better understanding of the challenges in the area that open up opportunities for future research. Also, it provides guidance for new studies. Possible future studies include defining and validating new models by considering the discussion and results presented throughout this work. In addition, a temporal update may also be performed in order to compare the evolution of this topic over time.

CRediT authorship contribution statement

Kellyton Brito: Conceptualization, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Rogério Luiz Cardoso Silva Filho:** Conceptualization, Investigation, Methodology, Writing – original draft. **Paulo Jorge Leitão Adeodato:** Conceptualization, Methodology, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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